

A 10x10 grid of dots on a black background. The dots are arranged in a regular pattern, forming a square grid. There are 10 dots in each row and 10 dots in each column, for a total of 100 dots.

Agenda

- ## Reference

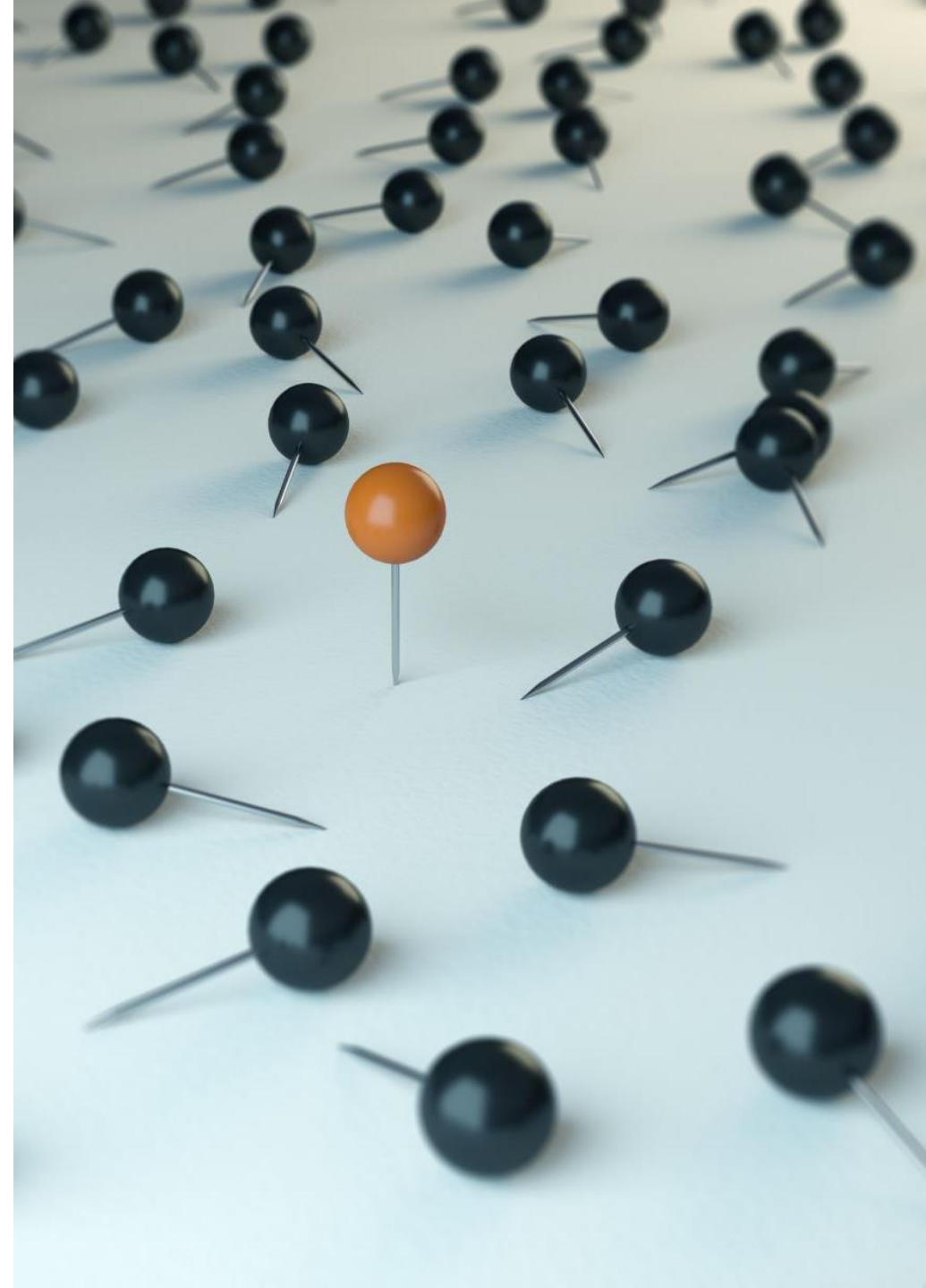
- Jarmul (2023) Practical Privacy Ch 1 (*optional*: Browse Ch 2, 4 and 8)

Data Governance

Framework for handling data

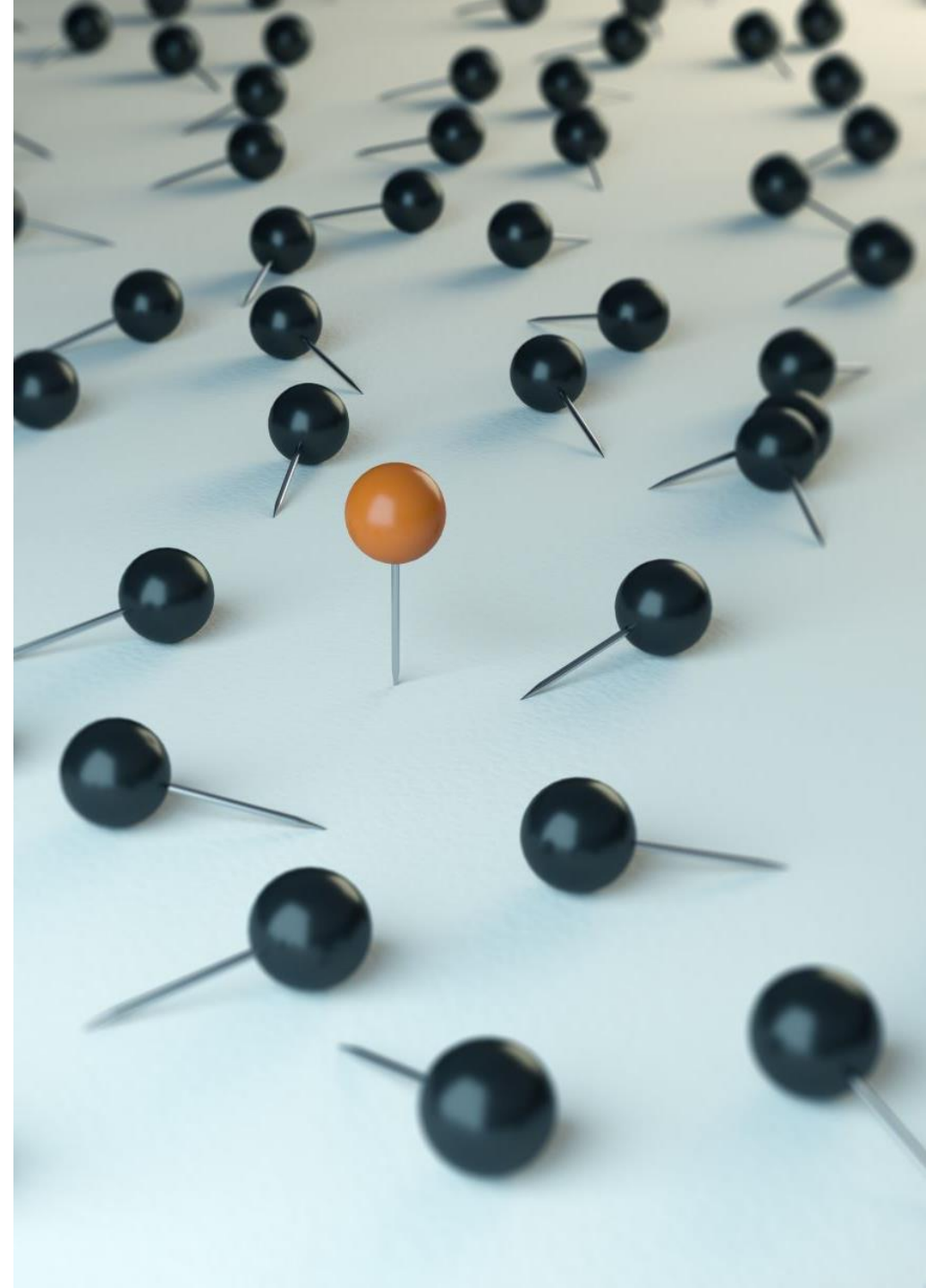
Managing:

- Whose data you collect
- How you collect it
- What you do with it after collection



Key questions to think about...

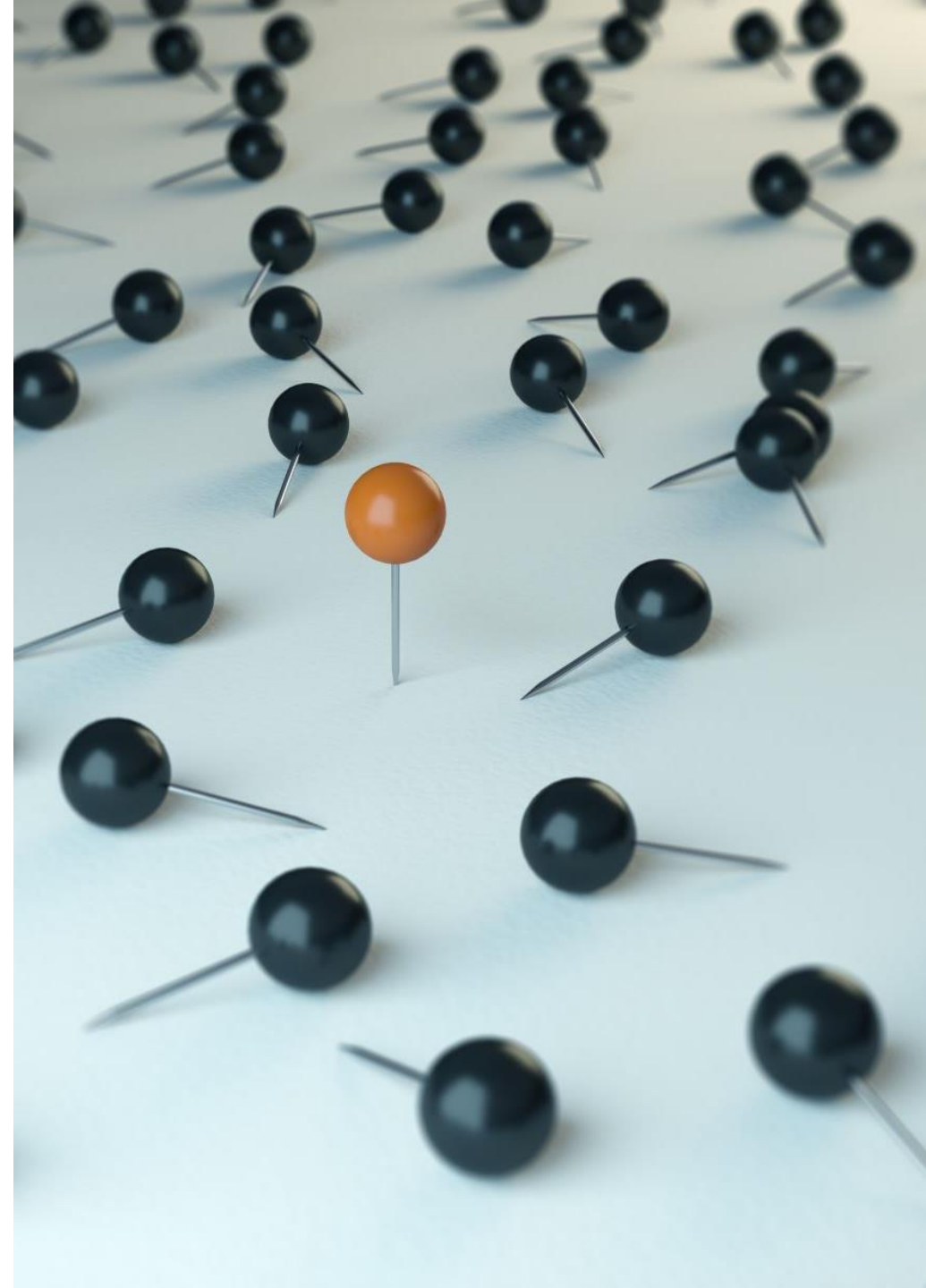
- When was this data collected? From where?
- How am I allowed to use this data? What consent was given at collection?
- What processing, cleaning, and preparation has this data undergone (i.e., removal of nulls, standardization of units, etc.)?
- Where else is this data processed, and where is related data stored?
- What should I keep in mind about the quality and origin of the data?



Sensitive Data

Person-related data (personally identifiable data)

- Name, gender, birth date
- Email, home address, IP address
- Social media profile, usernames
- Tax file number, licence number, credit card number, car registration, passport
- Health records, biometric data



Personally Identifiable Information (PII)

PII

Data that is unique or close to unique for a particular individual. This is usually defined in policy and regulations and can include things that you might not expect, such as your IP address, date of birth, and workplace.

Person-related data

Data that relates to a person but doesn't fall under PII. This could be anything related to their personhood, including interests, beliefs, locations, and online and offline behaviors and activities.

Proprietary and confidential data

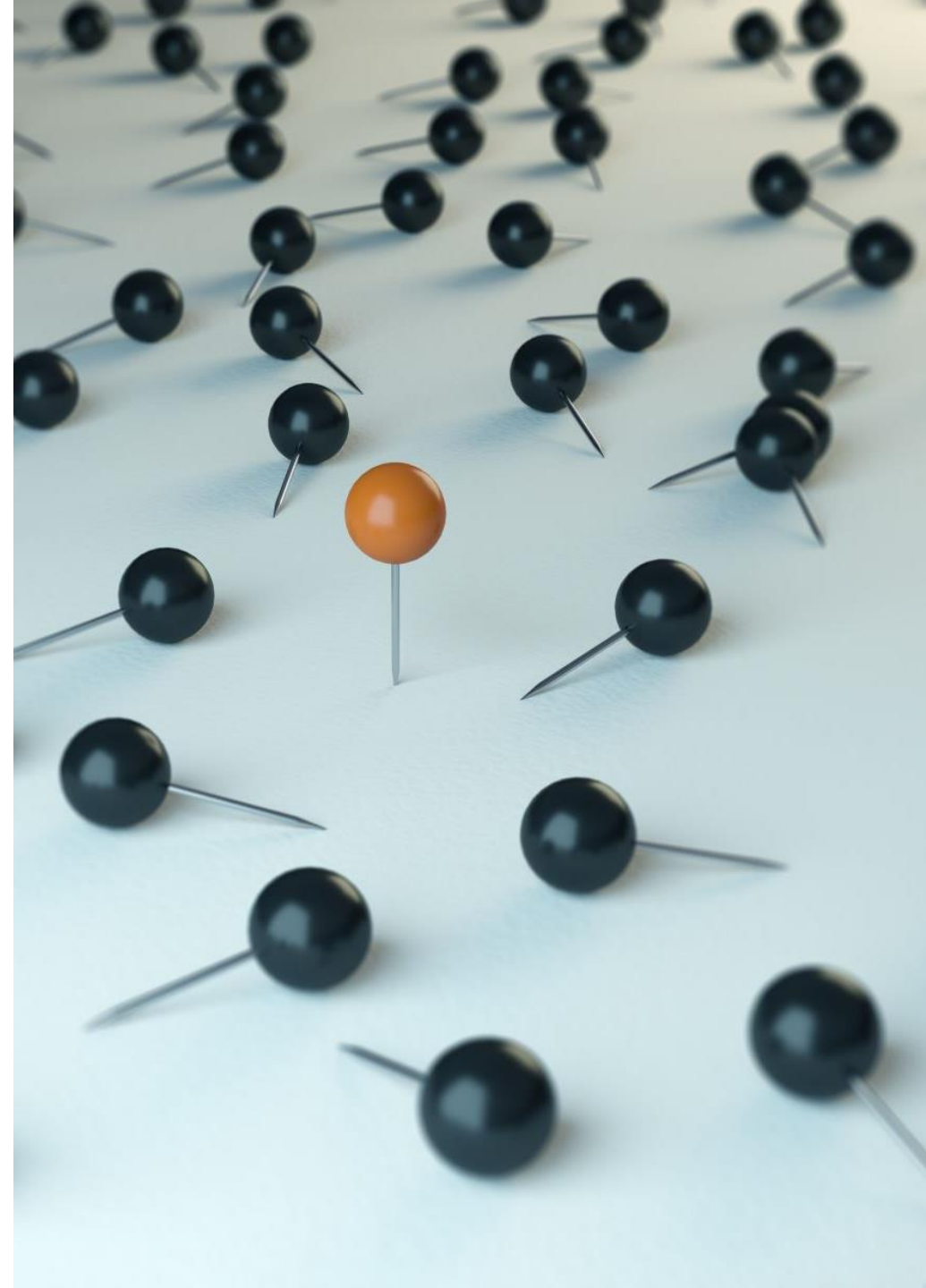
Data deemed sensitive for contractual or business-related purposes. Its release would endanger a business or other legal relationship or agreement.

Jarmul (2023) Ch1

Privacy Preferences

Different people different comfort level for different types of data

- Friendship groups
- Location
- Political affiliation, gender identity, religion

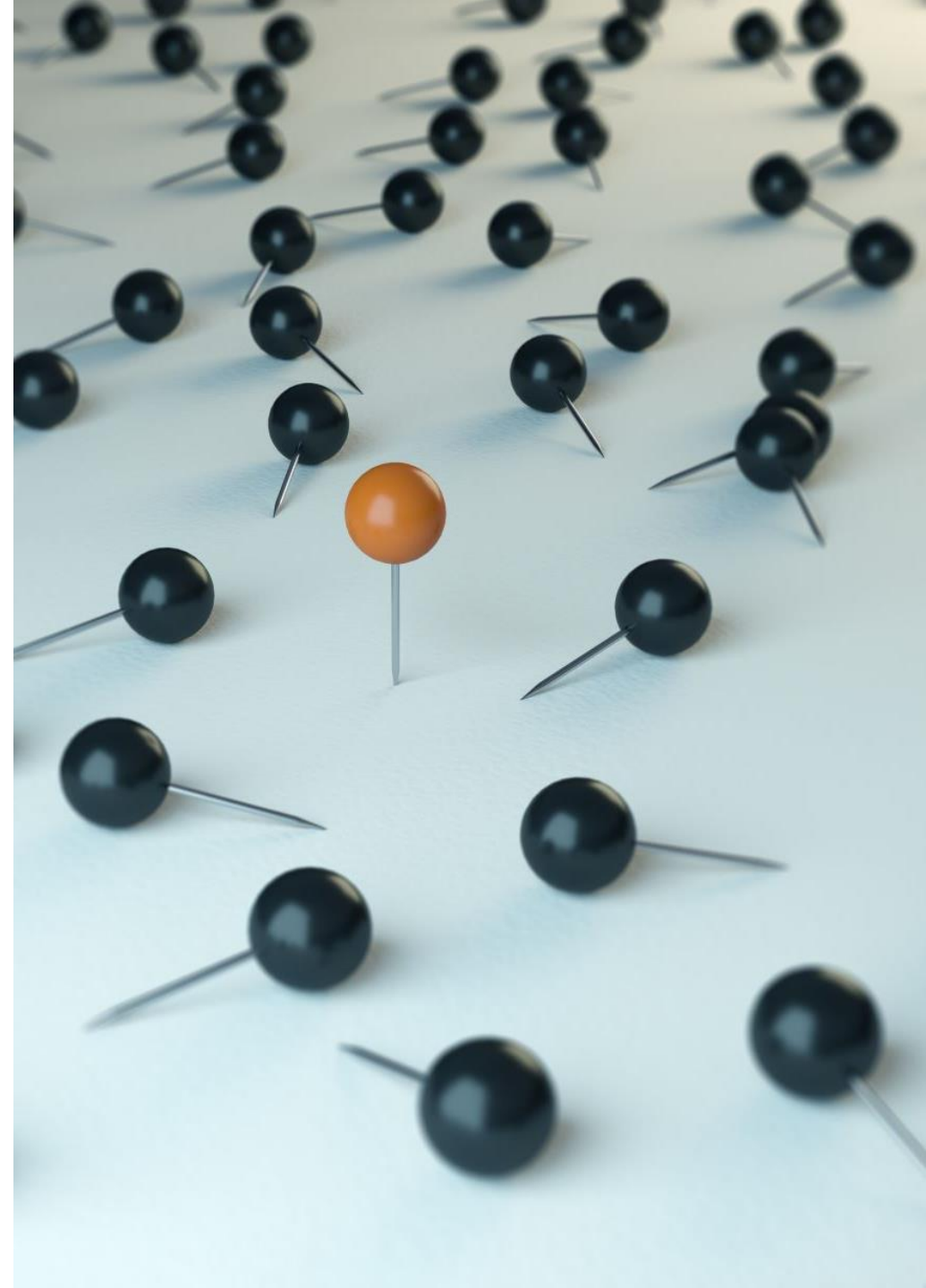


Sensitive data also includes...

Could be used when combined with other data to identify someone

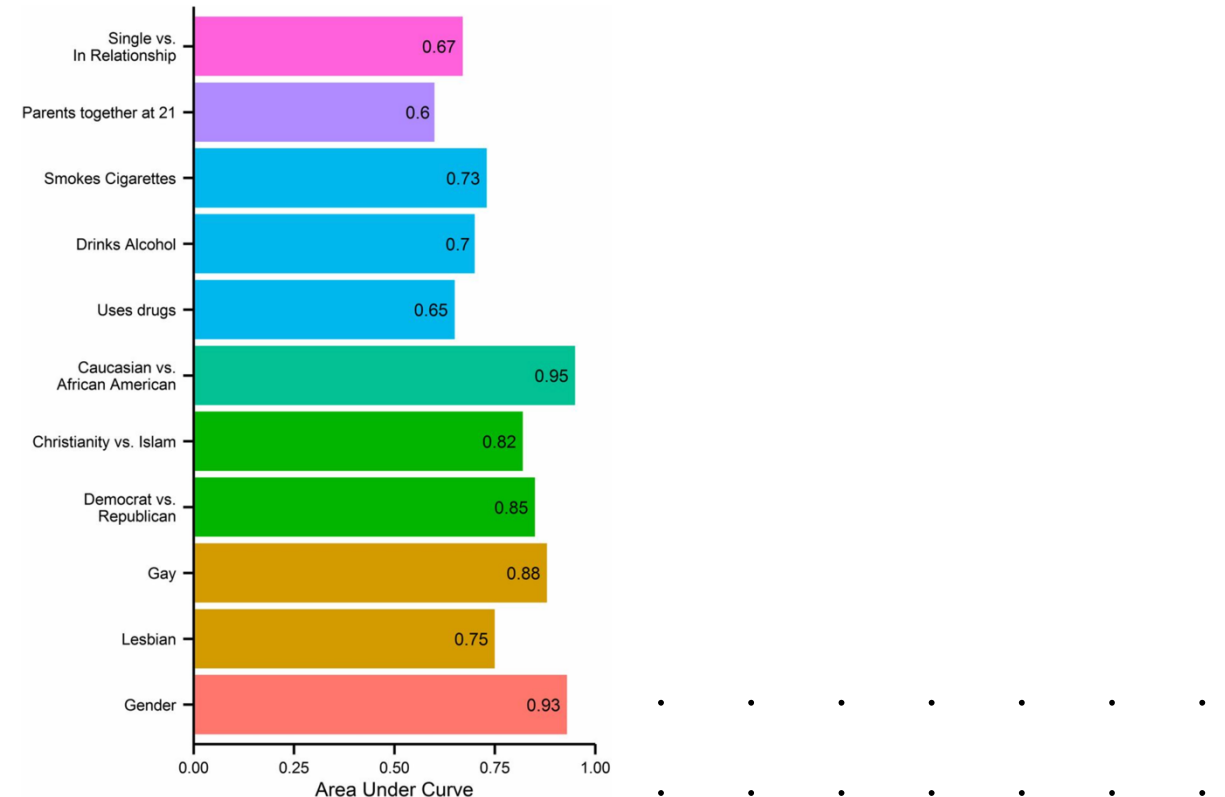
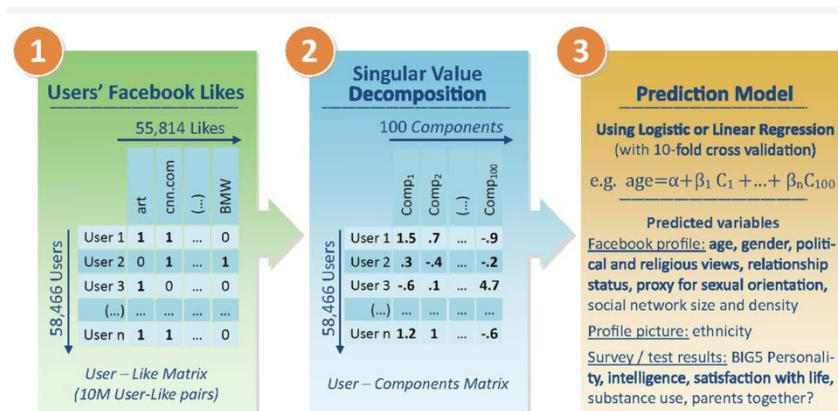
May not identify someone individually, but could be used to target them as part of a group

- Gambler
- Pregnant
- LGBTIQ+



What can you tell about a person about Facebook likes?

Gender, age, personality, religion, sexual and political orientation...



[Kosinski, M., Stillwell, D., & Graepel, T. \(2013\). Private traits and attributes are predictable from digital records of human behavior. *Proceedings of the national academy of sciences*, 110\(15\), 5802-5805.](#)

'Trade secrets'

Propriety or confidential non-person related data

For example, information about:

- Business
- Product

Data sharing

- Between parent and subsidiary company
- External entities (e.g., competitors, journalists, government/regulatory bodies...)

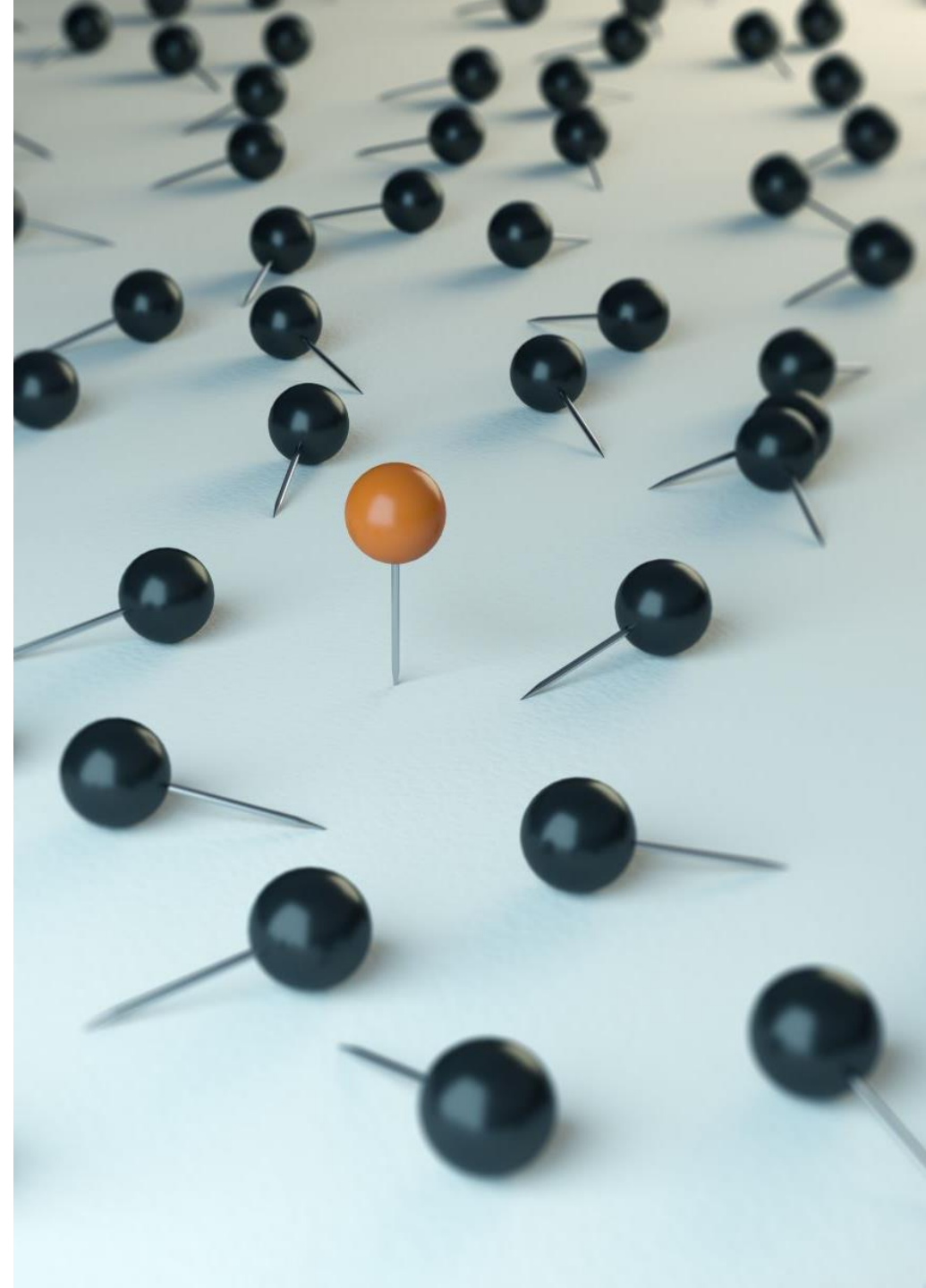
Whistleblowing: reporting
behaviour that may be
illegal or unethical

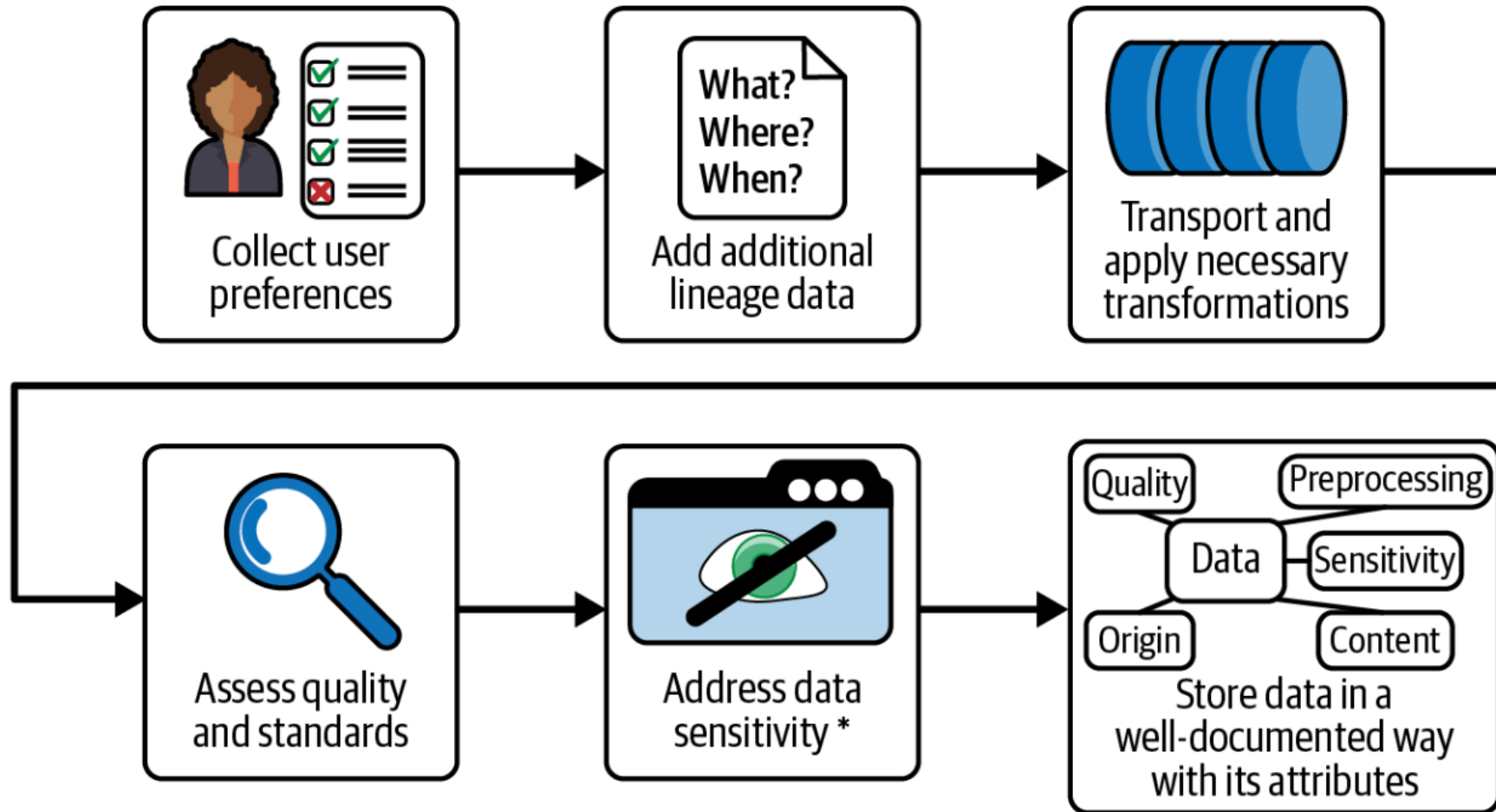
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Data documentation

When handling data in organisation consider recording information about:

- Data collection procedure
- Data quality
- Data security
- Data privacy
- Data definitions (e.g., data dictionary)





**Note: this step may need to be the first after collection if the architecture must guarantee certain privacy methods.*

Figure 1-2. Lineage and consent tracking pipeline

Anonymisation

Anonymise: *"to remove identifying information from (something, such as computer data) so that the original source cannot be known."*

Merriam-Webster Dictionary, accessed January 2025

Making it so source of data can't be known

Simple methods

- Removal of columns with sensitive data
- Aggregate row attributes
- Transformation

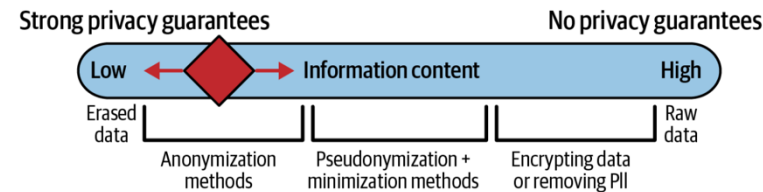
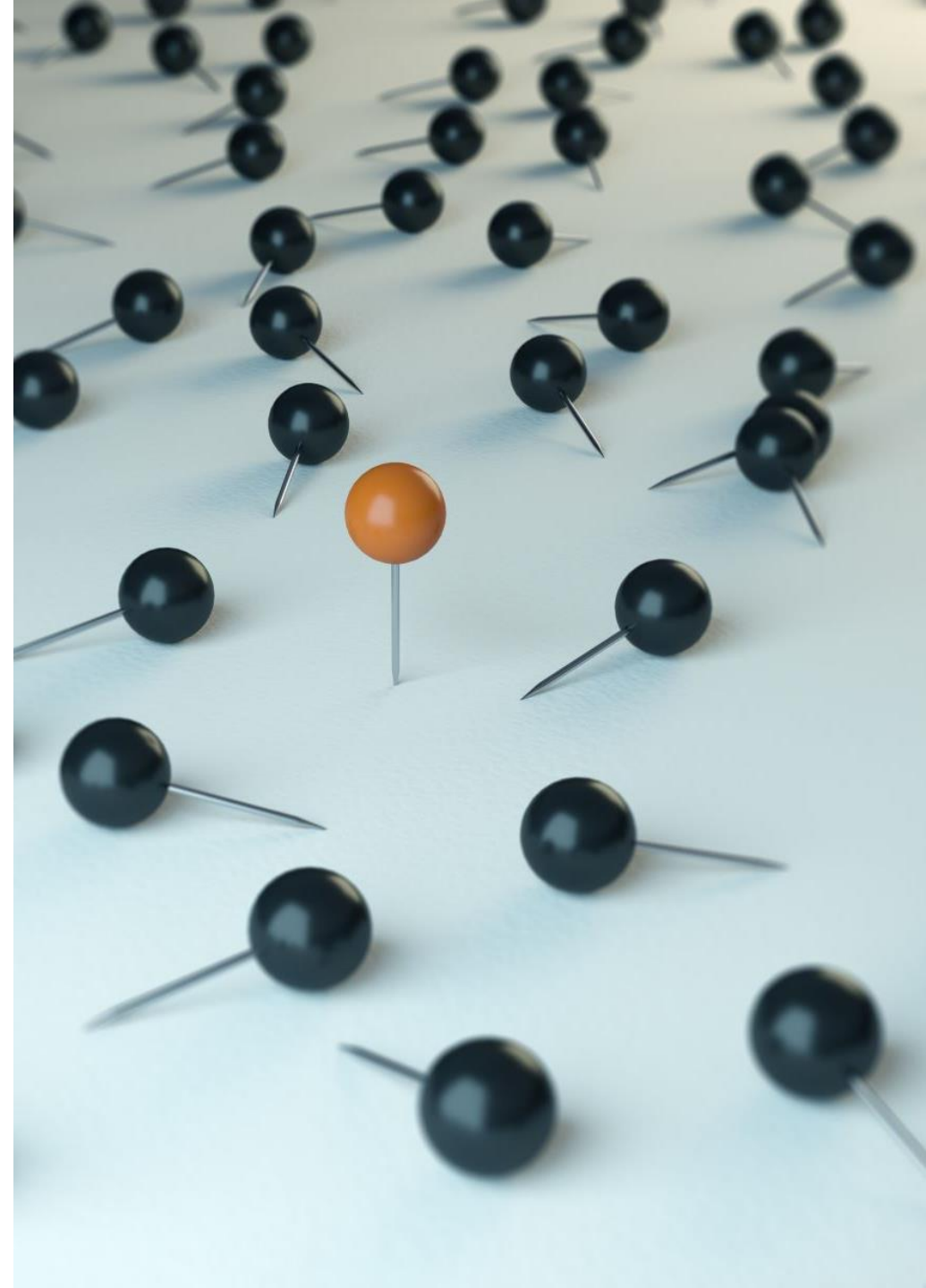


Figure 2-1. Information and privacy continuum



Anonymisation

Making it so source of data can't be known

Simple methods

- Removal of columns with sensitive data
- Aggregate row attributes
- Transformation

More complicated setting

- Encryption
- Pseudonymisation methods
- Anonymisation methods

What is your gender?
What is your age range?
Are you part or full time?
What department do you work in?

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Privacy Attacks

Netflix Prize Attack

Linkage Attack

Singling out Attack

Strava Heat Map Attack

Membership Inference Attack...



Legalities

General Data Protection Regulation ([GDPR](#))

Data right	Description
Right to be informed	People must be informed on how their data is being used, processed, collected, and so forth.
Right to access	People can access what information is held about them.
Right to rectification	People can correct information that is false or misleading.
Right to deletion/erasure	People can ensure companies delete their data.
Right to restrict processing	People can opt out or restrict how their data is being used and processed.
Right to data portability	People can take their data with them—to use themselves or to try competitor services.
Right to object	People can object to the usage of their data for particular uses (like marketing, research, and several other use cases).
Right to opt out of automated decision making	People can opt out of automated decision making processes, such as algorithmic or machine learning systems.



Other legal regulations to be aware of...

Australian Privacy Principles ([APP](#))

California Consumer Privacy Act ([CCPA](#))

Health ([HIPPA](#) - USA)



Company Policies

Important documents:

- Privacy Policies
- Terms of Service agreements

Check you are aware of policies that affect your work

Be prepared to work with legal professionals

- Understand your obligations
- Work out how to meet obligations

